

SAT Math

Area and Volume 1

Question #	ID		Answer
1.1	5252e606	The side length of a square is 55 centimeters (cm). What is the area, in cm^2, of the square? A. 110 B. 220 C. 3,025 D. 12,100	C
1.2	59cb654c	The area of a square is 64 square inches. What is the side length, in inches, of this square? A. 8 B. 16 C. 64 D. 128	A
1.3	0837c3b9	Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $DE = 2AB$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ? A. 10 B. 40 C. 80 D. 120	B
1.4	c88183f7	A rectangle has a length of 13 and a width of 6. What is the perimeter of the rectangle? A. 12 B. 26 C. 38 D. 52	C

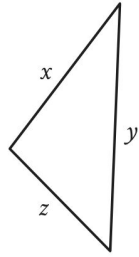
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Question # ID
1.5 29e9b28c

Answer

A



Note: Figure not drawn to scale.

The triangle shown has a perimeter of 22 units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

- A. 6
- B. 7
- C. 9
- D. 16

1.6 3453aafc

D

What is the area, in square centimeters, of a rectangle with a length of 36 centimeters and a width of 34 centimeters?

- A. 70
- B. 140
- C. 1,156
- D. 1,224

1.7 f60bb551

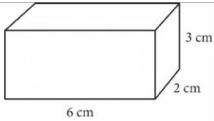
A

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560

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Question #	ID		Answer
1.8	4420e500	What is the area of a rectangle with a length of 4 centimeters (cm) and a width of 2 cm? A. 6 cm^2 B. 8 cm^2 C. 12 cm^2 D. 36 cm^2	B
1.9	165c30c4	A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?	2048
1.10	d683a9cc	The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh, where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?  The figure shows a right rectangular prism. The front horizontal edge is labeled 6 cm. The bottom receding edge is labeled 2 cm. The right vertical edge is labeled 3 cm. A. 36 B. 24 C. 12 D. 11	A
1.11	f39f88b7	A triangle has a base length of 40 centimeters and a height of 90 centimeters. What is the area, in square centimeters, of the triangle?	1800

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Question #	ID		Answer
1.12	93f48423	What is the area, in square inches, of a rectangle with a length of 7 inches and a width of 6 inches? A. 13 B. 20 C. 42 D. 84	C
1.13	d2047497	What is the area of a rectangle with a length of 17 centimeters (cm) and a width of 7 cm? A. 24 cm^2 B. 48 cm^2 C. 119 cm^2 D. 576 cm^2	C
1.14	a490003a	The width of a rectangle is 7 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle? A. 7 B. 14 C. 54 D. 329	D
1.15	50774285	Each base of a right rectangular prism has a length of 19 inches and a width of 8 inches. The prism has a volume of 2,736 cubic inches. What is the height, in inches, of the prism? A. 18 B. 27 C. 144 D. 152	A

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Question # ID
1.16 fc5ef8d3

Answer

C

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of $\overline{XY}$?

- A. 2
- B. 6
- C. 18
- D. 56

1.17 0e40dfb0

D

A rectangle has a length of 3 units and a width of 39 units. Which expression gives the area, in square units, of this rectangle?

- A. $2(3 + 39)$
- B. $2(3 \cdot 39)$
- C. $3 + 39$
- D. $3 \cdot 39$

1.18 2b41a4c4

880

A right rectangular prism has a length of 11 meters, a width of 8 meters, and a height of 10 meters. What is the volume, in cubic meters, of the prism?

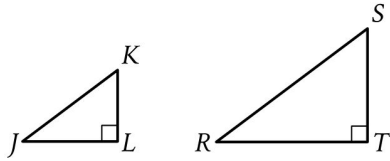
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Question # ID
1.19 babd7461

Answer

A



Note: Figure not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\overline{RS}$ is 135. What is the perimeter of triangle RST ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

1.20 bb560789

D

Triangle R has an area of 80 square centimeters (cm^2). Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2 ?

- A. 42
- B. 44
- C. 84
- D. 96